



What is IANA?

◆ Full Form:

Internet Assigned Numbers Authority



What is IANA?

IANA is a global organization that **manages and distributes critical internet resources** like:

- 1. IP addresses** (IPv4 & IPv6 blocks)
- 2. DNS root zone** (e.g., .com, .org, etc.)
- 3. Protocol numbers and ports** (e.g., HTTP = port 80)



Role:

IANA does not assign IPs directly to individuals or companies. It **allocates large blocks of IPs** to the **RIRs (Regional Internet Registries)**, which then manage and distribute them further.



What is RIR?

◆ Full Form:

Regional Internet Registry



What is an RIR?

RIRs are **regional organizations** responsible for **managing IP address allocation** and **autonomous system numbers (ASN)** in specific areas of the world.

They **get IP address blocks from IANA** and **assign them to ISPs, governments, and organizations** in their regions.



List of 5 RIRs:

RIR Name	Region Covered	Headquarters
ARIN	North America	USA
RIPE NCC	Europe, Middle East, Central Asia	Netherlands
APNIC	Asia Pacific	Australia
LACNIC	Latin America, Caribbean	Uruguay

Flow of IP Address Allocation

IANA (Global)



RIR (Regional)



ISP / LIR (Local Internet Registry)



End Users / Companies / Servers

Example:

- IANA gives 43.0.0.0/8 block to **APNIC**
- APNIC gives a smaller block to **Airtel (India)**
- Airtel assigns IP 43.1.2.3 to your **home router**

Why Are They Important?

IANA

Manages global IP blocks

Maintains global DNS root

Ensures fairness across globe

RIR

Distributes regionally

Helps assign ASNs & IPs

Supports local policies

Who Oversees IANA?

- IANA is operated by **ICANN** (Internet Corporation for Assigned Names and Numbers).
- ICANN ensures **no country or company controls the internet** alone — it stays decentralized.

Real-World Analogy:

Imagine **IANA** as a **central water authority**.

- It gives large water supplies to **regional plants (RIRs)**.
- Then the **RIRs** distribute it to **cities, then homes and industries**.